

Python Lab Questions – 4th June 2026

1. Prime Number Analyzer

Problem Statement:

Accept a number from the user and determine whether it is a prime number or not.

Additional Requirement:

If the number is not prime, display all its factors.

Example:

Input: 15

Output:

Factors: 1 3 5 15

15 is not a Prime Number

2. Armstrong Number Checker

Problem Statement:

Accept a number from the user and check whether it is an Armstrong Number.

Example:

Input: 153

Output:

153 is an Armstrong Number

3. Strong Number Verification

Problem Statement:

A Strong Number is a number whose sum of factorials of digits equals the number itself.

Write a program to check whether a given number is a Strong Number.

Example:

Input: 145

Output:

145 is a Strong Number

4. Number Guessing Game

Problem Statement:

Generate a secret number between 1 and 50.

Allow the user to keep guessing until the correct number is found.

Display:

- "Too High"
- "Too Low"
- "Correct Guess"

Also display the total number of attempts.

5. Electricity Bill Calculator

Problem Statement:

Calculate electricity bill based on the following slab rates:

Units	Rate
0-100	₹5/unit
101-200	₹7/unit
Above 200	₹10/unit

Display:

- Units Consumed
- Total Bill
- Category (Low / Medium / High Consumption)

6. Student Result Management System

Problem Statement:

Accept marks of 5 subjects.

Display:

- Total Marks
- Percentage
- Grade

Grade Criteria:

Percentage Grade

>=90	A+
>=75	A
>=60	B
>=40	C
<40	Fail

Also display the number of subjects failed.

7. Pattern-Based Number Pyramid

Problem Statement:

Accept the number of rows and print the following pattern:

For Input:

5

Output:

1
12
123
1234
12345

Challenge:

Print the reverse pattern as well.

8. ATM Simulation System

Problem Statement:

Initial Balance = ₹10,000

Display a menu repeatedly:

1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Requirements:

- Withdrawal should not exceed balance.
 - Display appropriate messages.
 - Continue until Exit is selected.
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9. Palindrome and Reverse Number Checker

Problem Statement:

Accept a number from the user.

Display:

- Reverse Number
- Whether it is a Palindrome

Example:

Input: 1221

Output:

Reverse: 1221

Palindrome Number

10. Mini Employee Payroll System

Problem Statement:

Accept:

- Employee Name
- Basic Salary

Calculate:

Component	Percentage
HRA	20%
DA	10%
PF Deduction	12%

Display:

- Gross Salary
- Net Salary

Additionally:

Net Salary > 50000 → Senior Grade
Net Salary > 30000 → Mid Grade
Else → Junior Grade